INVOLUTION AND COMMUTATOR-WIDTH OBSTRUCTIONS ON UNRESTRICTED THOMPSON'S GROUP F

ARGUS OPEN RESEARCH PROJECT

Abstract. Problem 3.1 of the AIM problem list on Thompson's groups includes F alongside T, V, and topological full groups when asking to prove that every element is a product of at most three involutions ($\rho_2(g) \le 3$) and has commutator width one ($\rho_1(g) = 1$). We record the exact elementary obstructions on unrestricted F, witnessed by the explicit dyadic piecewise-linear homeomorphism $f \in F$ with breakpoints $(0, 1/2, 3/4, 1)$ and slopes $(1/2, 1, 2)$.

1. Main Proposition

Let F be Thompson's group of orientation-preserving piecewise-linear dyadic homeomorphisms of $[0, 1]$. Define the endpoint slope homomorphism

$$\chi : F \longrightarrow \mathbb{Z}^2, \qquad \chi(g) \;=\; \big(\log_2 g'_+(0),\ \log_2 g'_-(1)\big).$$

Theorem 1.1. *Let $f \in F$ be defined by*

$$f(x) \;=\; \begin{cases} \frac{1}{2}x, & 0 \le x \le \frac{1}{2}, \\ x - \frac{1}{4}, & \frac{1}{2} \le x \le \frac{3}{4}, \\ 2x - 1, & \frac{3}{4} \le x \le 1. \end{cases}$$

Then $f \neq \mathrm{id}$, $f(1/2) = 1/4$, $\chi(f) = (-1, 1)$, and:

(1) *Every involution $h \in F$ ($h^2 = \mathrm{id}$) is equal to* id *because h is strictly increasing on $[0, 1]$; hence f is not a finite product of involutions in F.*
(2) *Since $\mathbb{Z}^2$ is abelian, $\chi([a, b]) = (0, 0)$ for all $a, b \in F$, so χ vanishes identically on the derived subgroup $[F, F]$. Because $\chi(f) = (-1, 1) \neq (0, 0)$, f is not a finite product of commutators in F.*

Remark 1.2. This resolves only the literal unrestricted inclusion of F in the problem wording; commutator width restricted to the simple derived subgroup $[F, F]$ and the assertions on the simple groups T and V remain separate.

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 20001352 in the domain of geometric group theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.
(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.
(c) *Classification:* Within the program taxonomy, this result is categorized as a *literal scope counterexample* deciding the scope: literal unrestricted inclusion of Thompson's group F in the involution-product and commutator-width assertions.

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3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- commutator width restricted to [F,F].
- simple Thompson groups T and V.

References

[1] Primary problem list and source literature for Result 20001352 (geometric group theory), archived with exact theorem locators and verification certificates in the accompanying repository package.